



FIRE: THE FIRST-YEAR INNOVATION & RESEARCH EXPERIENCE

QUANTUM MACHINE LEARNING

The Power of Machine Learning A Deep Dive into LHC Data

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Introduction - The Large Hadron Collider

The Large Hadron Collider (LHC) is the world's largest and most powerful particle accelerator. It was built to investigate the fundamental nature of matter and the origins of the universe by colliding subatomic particles at incredibly high energies. With a circumference of 27 kilometers, the LHC is an engineering marvel, consisting of superconducting magnets and advanced detectors. Since its first operation in 2008, the LHC has made groundbreaking discoveries, including the Higgs boson, and continues to push the boundaries of our understanding of the universe.

We employ Deep Learning to analyze the extensive data produced by LHC experiments, helping us identify rare events and make predictions. Combining LHC with machine learning aims to unveil more cosmic mysteries and advance our understanding of fundamental matter.

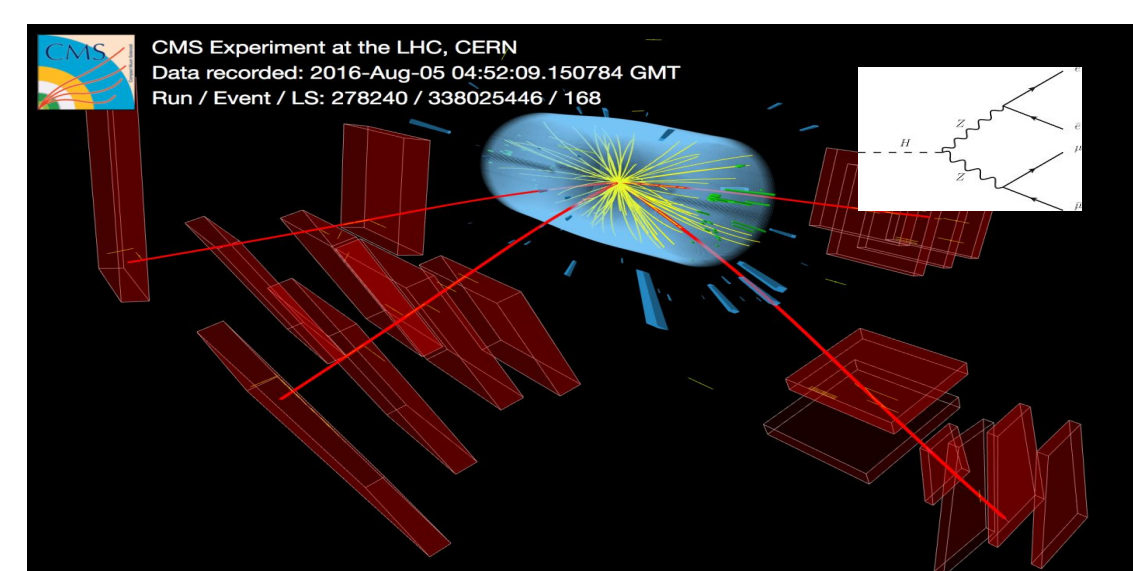


Figure 1. A Higgs boson candidate transforming into four muons (red lines). (Image: CERN)

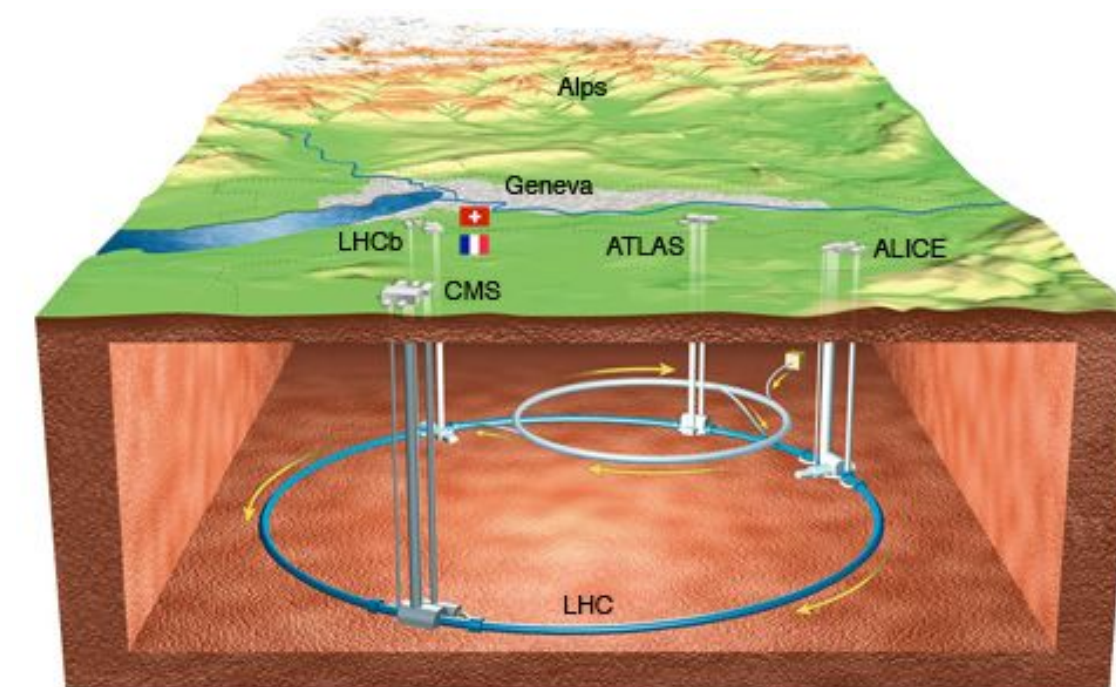


Figure 2. The Large Hadron Collider at CERN, located near Geneva, Switzerland.

What is Universe Made of?

The matter universe, as we know it, is composed of different types of particles, including quarks, leptons, and force carrying particles. These particles interact with each other through fundamental forces, such as the electromagnetic, weak, and strong forces. However, there are still many unanswered questions about the nature of matter and the fundamental forces that govern it. This is where the Large Hadron Collider (LHC) comes in.

The LHC is a particle accelerator that allows scientists to recreate the conditions that existed right after the Big Bang, and study the behavior of particles at high energies. By colliding particles at nearly the speed of light, the LHC allows physicists to probe the fundamental structure of matter and search for new particles that could help explain the mysteries of the universe, such as dark matter and dark energy. The discoveries made at the LHC have the potential to revolutionize our understanding of the universe and the nature of matter itself.

Tre generationer av materia (fermioner)									
	I	II	III	IV	V	VI	VII	VIII	IX
massa	2.4 MeV/c ²	1.27 GeV/c ²	171.2 GeV/c ²	0	0	1 GeV/c ²	0	0	0
laddning	2/3	2/3	2/3	0	0	0	0	0	0
spin	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2
namn	u	c	t	b	g	u	d	s	b
	upp	charm	topp	bottom	gluon	upp	down	strange	bottom
Kvarkar	4.8 MeV/c ²	1.30 MeV/c ²	171.2 GeV/c ²	4.2 GeV/c ²	0	1.2 GeV/c ²	0.5 MeV/c ²	0.1 MeV/c ²	1.7 MeV/c ²
	2/3	2/3	2/3	2/3	0	2/3	1/3	1/3	1/3
	1/2	1/2	1/2	1/2	1	1/2	1/2	1/2	1/2
	d	s	b	g	0	u	d	s	b
	down	strange	bottom	gluon		up	down	strange	bottom
Leptoner	0.511 MeV/c ²	105.7 MeV/c ²	1.777 GeV/c ²	80.4 GeV/c ²	0	0.511 MeV/c ²	105.7 MeV/c ²	1.777 GeV/c ²	80.4 GeV/c ²
	1/6	1/6	1/6	1/6	0	1/6	1/6	1/6	1/6
	1/2	1/2	1/2	1/2	1	1/2	1/2	1/2	1/2
	e	μ	τ	W	0	e	μ	τ	W
	electron	muon	tau lepton	W boson		electron	muon	tau lepton	W boson
Gaugebosoner	0.511 MeV/c ²	105.7 MeV/c ²	1.777 GeV/c ²	80.4 GeV/c ²	0	0.511 MeV/c ²	105.7 MeV/c ²	1.777 GeV/c ²	80.4 GeV/c ²
	1/6	1/6	1/6	1/6	0	1/6	1/6	1/6	1/6
	1/2	1/2	1/2	1/2	1	1/2	1/2	1/2	1/2
	e	μ	τ	W	0	e	μ	τ	W
	electron	muon	tau lepton	W boson		electron	muon	tau lepton	W boson